

Problem Set 5: Web Scraping and APIs

Jhinuk Banerjee

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1 Web Scraping: Perfume Data from Wikipedia

What data did I collect?

I scraped a table for comparing the natural versus the synthetic perfume ingredients from a Wikipedia article on Perfumes (<https://en.wikipedia.org/wiki/Perfume>). This table contains five rows comparing natural and synthetic perfumes across different dimensions like variance, components, scent uniqueness, scent complexity, and price. I used the `rvest` and `httr` packages in R to send an HTTP GET request with a browser user-agent header, parse the returned HTML, and extract the `wikitable`-class table into a data frame.

Why does this interest me?

I really like perfumes and thought it would be interesting to get data on something I like. The natural versus synthetic divide in the fragrance industry reflects different tensions in luxury markets, like authenticity, relevance for consumption and gender-targeted marketing. The price row is interesting from an economics perspective, as it reflects how the extraction methods and supply chain constraints shape luxury goods pricing.

Will it be useful?

This table is quite illustrative, and the scraping skills practiced here can be used to collect larger fragrance datasets. For instance, getting cross-sectional data on perfume characteristics, gender targeting and pricing could be useful for future research on gender and consumption.

Resources used

I used the `rvest` tidyverse documentation (<https://rvest.tidyverse.org/>) and the `httr` package documentation for guidance on setting user-agent headers to handle HTTP connections reliably.

2 API Data: World Bank Gender Indicators

What data did I collect?

I used the `wbstats` package in R to query the World Bank API. I retrieved two indicators for all countries from 2000 to 2023: (1) female labor force participation as a share of the total labor force (`SL.TLF.TOTL.FE.ZS`), and (2) the share of seats held by women in national parliaments (`SG.GEN.PARL.ZS`). The resulting dataset contains 5,208 country-year observations.

What did I find?

The data reveal significant cross-national variation in both indicators. Female labor force participation ranges from a minimum of about 6.8% to a maximum of 54.8%, with a mean of 41.1%. Women's parliamentary representation ranges from 0% to 63.75%, with a mean of 18.8%, reflecting that women remain underrepresented in legislatures globally, even though their participation has grown over the period.

Why is this useful?

These indicators are directly relevant to my dissertation research on economic insecurity and gender in politics. The World Bank panel provides clean, standardized cross-national data that could complement my U.S. state legislature elections data.

Packages used

I used the `wbstats` package (v1.1), which provides a clean R interface to the World Bank's public API. No API key is required. Documentation is available at <https://CRAN.R-project.org/package=wbstats>.